Segment- and Track-level Validation of the `lhcb_velo_toy` Model: A Hopfield-Hamiltonian Reconstruction Stress Test up to 10^3 Tracks per Event

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Abstract

We report a comprehensive validation of the `lhcb_velo_toy` simulation and reconstruction stack — a five-plane VeLo-like toy detector coupled to a Hopfield-style Hamiltonian segment solver — against an earlier reference implementation, and extend the characterisation regime to $n_{\text{tracks}} = 1000$ tracks per event. Using an analytically motivated angular acceptance $\varepsilon(\sigma_s, \sigma_r)$ we verify that (i) the segment-matching efficiency stays pinned at 100% across the full range of physical noise and track multiplicity, (ii) the classical Hamiltonian solver delivers 100% segment efficiency up to $n_{\text{tracks}} = 1000$ with false activation rate rising from 0 at low multiplicity to $\approx 20\%$ at $n_{\text{tracks}} = 1000$, (iii) the matrix spectrum remains strictly positive ($\lambda_{\min} \geq 0.88$, condition number $\kappa \leq 9$), and (iv) the dominant failure mode at high n_{tracks} is *post-processing* — the default connected-components tracker merges false-segment bridges into mega-clusters and collapses reconstruction efficiency from 99% to 43% between $n_{\text{tracks}} = 300$ and $n_{\text{tracks}} = 1000$, whereas a layer-exclusive tracker restores efficiency to $\gtrsim 99.9\%$ at the cost of a 23.6% ghost rate. The validated operating point for the Hamiltonian segment solver is $(\gamma, \delta, \varepsilon, \tau) = (3,\ 1,\ 2\,\text{mrad},\ 0.35)$.

1 Introduction

The `lhcb_velo_toy` package is a lightweight VeLo-like detector simulator and reconstruction framework used to prototype Hopfield-style [1–4] and quantum [5, 6] track-finding algorithms on realistic multiplicities. This report addresses four specific questions regarding its correctness and operating envelope:

1. Does the analytic angular threshold $\varepsilon(\sigma_s, \sigma_r)$ faithfully match the true-segment angle distribution under realistic multiple-scattering and hit-resolution noise?

2. Does the package reproduce the segment-level acceptance curves of the older reference notebook `track_density_test_local.ipynb` when parameters are matched?

3. How does the classical Hopfield Hamiltonian solver scale up to $n_{\text{tracks}} = 1000$, in terms of (a) segment efficiency, (b) false-rate survival, (c) spectral conditioning, and (d) track-level reconstruction metrics?

4. Where does the stack fail first, and is that failure in the solver, the acceptance model, or the post-processing tracker?

The analysis is carried out in a single master notebook `segment_level_analysis.ipynb` with 17 sections, whose structure is mirrored in this report. All figures referenced below are cached artefacts in `outputs/segment_analysis/` and were not regenerated for this write-up.

2 Model and methodology

2.1 Detector and event generator

The detector consists of five parallel plane modules spaced by $\Delta z = 33$ mm, centred on $z \in \{33, 66, 99, 132, 165\}$ mm, each of active area 80×80 mm^2 (half-width 40 mm) except for the §3.4 reference-reproduction experiment, which uses 160×160 mm^2 for parameter matching to the earlier study. Primary vertices are drawn from a Gaussian with $\sigma_{xyz} = (0, 0, 1)$ mm in the default sweeps and $(1, 1, 1)$ mm in the reference-reproduction experiment. Tracks are generated by `lhcb_velo_toy.generation.StateEventGenerator` with uniform $\phi \in [-\phi_{\max}, \phi_{\max}]$ and identical θ-cone. Hit positions are perturbed by a Gaussian measurement error σ_r; the propagation direction is perturbed at each module by a multiple-scattering kick σ_s.

2.2 Angular acceptance ε

A segment *pair* sharing a common middle hit is accepted if the opening angle between the two segments is below a threshold ε computed as

$$\varepsilon(\sigma_s, \sigma_r) \;=\; \sqrt{2\,\theta_s^2 \;+\; 12\,\theta_r^2 \;+\; 2\,\theta_{\min}^2}\,, \tag{1}$$

with $\theta_s = s\,\sigma_s$, $\theta_r = \arctan(s\,\sigma_r/\Delta z)$, $s = 3$, and $\theta_{\min} = 1.5 \times 10^{-5}$ rad. The factor 12 on the hit-resolution term propagates the kink due to independent resolution errors on both segments through the triplet. The $\theta_{\min}^2$ floor guarantees a non-zero threshold even in the zero-noise limit.

2.3 Segment-pair definitions

Two definitions are used in parallel:

1. *Shared-middle-hit (triplet)*: all ordered triples of hits (h_p, h_m, h_n) on consecutive modules. A pair is *true* iff both component segments belong to the same truth track. Combinatorics $\mathcal{O}(T^3)$.
2. *Pairwise (truth segments only)*: one segment per truth-track consecutive-hit pair, all $\binom{N_{\text{seg}}}{2}$ pairs enumerated; true iff both segments are from the same truth track. Combinatorics $\mathcal{O}(T^2)$.

The two methods share the true-pair population exactly but differ by orders of magnitude in the false-pair pool and thus in the predicted false-acceptance rate. All sweeps are evaluated in both methods and cross-checked against pure-Python reference kernels before using the Numba-JIT [7] accelerated versions for large multiplicities.

2.4 Hopfield Hamiltonian and solver

The segment Hamiltonian $H(\gamma, \delta)$ is constructed by `SimpleHamiltonianFast` with penalty $\gamma = 3$ and bias $\delta = 1$. Its off-diagonal entries reward angle-consistent same-module-chain segment pairs and penalise hit-sharing conflicts; the diagonal is $(\delta + \gamma)$ so that for an isolated segment the Hopfield fixed point is $s^* = \delta/(\delta + \gamma) = 0.25$. The full system $Hs = b$ is solved by direct sparse Cholesky (SciPy); a threshold $\tau = 0.35$ separates active segments ($s_i \geq \tau$) from inactive ones. For an isolated five-hit truth chain the outermost-segment fixed point is the geometric invariant $s^*_{\text{outer}} = 0.3636\ldots$ which is used as a solver health-check throughout.

2.5 Trackers

Two post-processors are compared: `get_tracks` assembles connected components from the set of active segments, and `get_tracks_layered` is a module-exclusive greedy variant seeded at the first module that enforces angular consistency between consecutive segments.

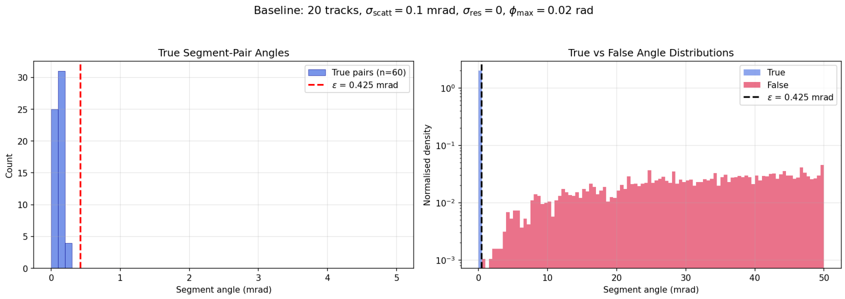

Figure 1: Pair-angle distributions for a baseline event (20 tracks, $\sigma_s = 0.1$ mrad, $\sigma_r = 0$). Dashed line: analytic ε from Eq. (1). Source: segment_level_analysis.ipynb §3.

2.6 Validation

Reconstructed tracks are matched against the truth with `EventValidator.match_tracks` using LHCb-standard cuts (purity ≥ 0.70, `min_rec_hits` $= 3$) [8].

2.7 Implementation checks

All Numba kernels are byte-validated against pure-Python references before deployment; the custom vectorised Hamiltonian builder used in §4 is *exactly* (matrix- and vector-equal) the same assembly as `SimpleHamiltonianFast.construct_hamiltonian` [9].

3 Segment-level results

3.1 Baseline (§3, §4)

Figure 1 shows the true- and false-pair angle distributions for a single event of 20 tracks at $\sigma_s = 0.1$ mrad, $\sigma_r = 0$. The true peak is confined well below the analytic ε; the false distribution has bulk density at tens of mrad. The scattering-sweep histograms confirm that ε tracks the mean of the true-angle distribution across $\sigma_s \in [0.05, 1.0]$ mrad and keeps the true-pair acceptance above 80% while leaving the false tail unaffected.

3.2 Scattering and resolution sweeps (§5, §6)

For $n_{\rm tracks} \in \{10, 20, 50, 100\}$, $\sigma_s \in \{0.05, \ldots, 1.0\}$ mrad and $\sigma_r \in \{0, 5, \ldots, 50\}$ μm, the segment efficiency (true-pair acceptance) is pinned at 100% across both angle cones $\phi_{\max} \in \{0.02, 0.2\}$ (Fig. 2). The false-rate climbs monotonically with noise, as expected because ε widens. The resolution term dominates above $\sigma_r \gtrsim 20$ μm as anticipated from the $12\theta_r^2$ coefficient in Eq. (1).

3.3 Density scan and ROC (§7, §8)

Figure 3 plots segment efficiency, false rate, and raw true/false pair counts versus $n_{\rm tracks}$ at fixed noise. True pairs scale as $n_{\rm tracks}$; false pairs as $n^3_{\rm tracks}$ (triplet method), confirmed by the reference n^3 line in panel (d). The ROC acceptance curves (not shown) place the analytic ε operating point on the knee between $> 98\%$ true-acceptance and $\lesssim 1\%$ false-acceptance for all tested $\sigma_s \leq 0.4$ mrad.

3.4 Fixed-ε comparison, new vs. old (§10, §10b, §10c)

To reproduce the earlier reference notebook `track_density_test_local.ipynb`, the experiment is re-run with identical generator parameters (MIP, PV $\sigma = (1, 1, 1)$ mm, 160×160 mm^2 modules, $\sigma_s = 0.1$ mrad, $\sigma_r = 5$ μm) but under *both* segment-pair definitions of §2, at a fixed $\varepsilon = 2$ mrad.

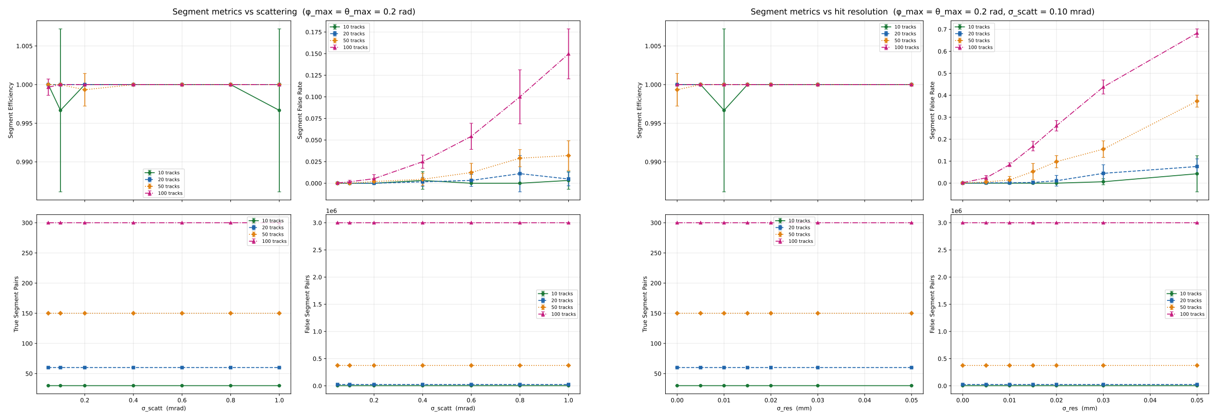

Figure 2: Scattering (left) and resolution (right) sweeps at $\phi_{\text{max}} = 0.2$. Segment efficiency stays at 100%; false rate grows monotonically with noise.

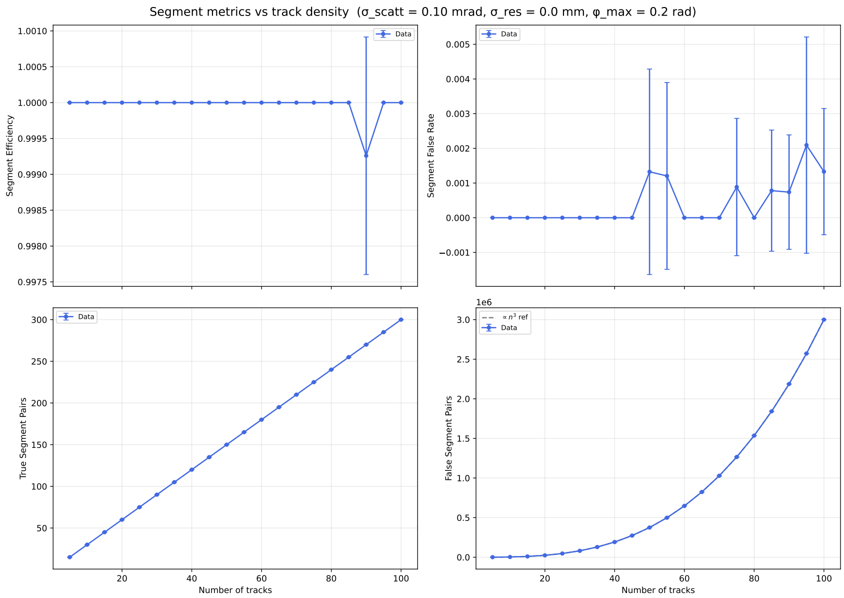

Figure 3: Density scan, $n_{\text{tracks}} = 5 \ldots 100$, $\sigma_s = 0.1$ mrad, $\sigma_r = 0$. Panel (d) includes a reference n^3 slope.

Segment efficiency is 100% for both methods and at every multiplicity. False rates diverge: the triplet method produces a false pool growing as T^3, the pairwise method as T^2. A weighted power-law fit $N = aT^b$ on the false-pair counts recovers $b \approx 3$ (triplet) and $b \approx 2$ (pairwise), matching the combinatorial expectation (Fig. 4).

3.5 Zero-noise benchmark (§11)

At $\sigma_s \to 0$, $\sigma_r = 0$, ε collapses to the floor $\sqrt{2}\,\theta_{\min} \approx 0.0212\ \mu$rad. Segment efficiency remains $\geq 95\%$ in both the dense ($\phi_{\max} = 0.02$) and non-dense ($\phi_{\max} = 0.2$) regimes. The dense configuration shows a higher residual false-rate floor due to coincidental alignment of small-angle tracks.

3.6 Extended range to 10^3 tracks (§12, §13)

Re-running §5–§7 and §11 with n_{tracks} extended to $\{200, 500, 1000\}$ (Numba-JIT triplet kernel) confirms all the scaling behaviours observed at lower multiplicity. Segment efficiency stays

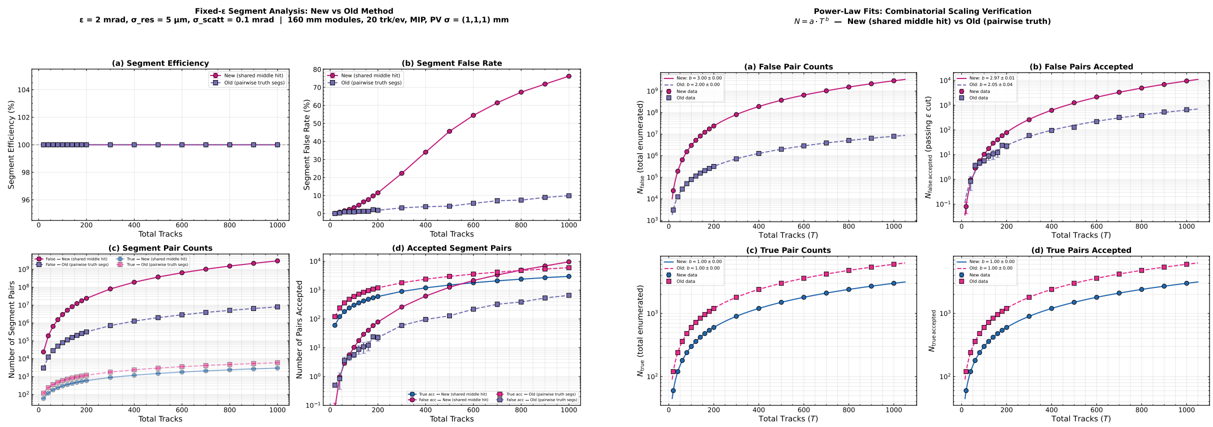

Figure 4: Left: fixed-$\varepsilon = 2$ mrad efficiency/false-rate comparison between triplet (new) and pairwise (old-reference) metrics. Right: power-law fits of false- and true-pair counts vs. multiplicity, recovering the $b \approx 3$ (triplet) and $b \approx 2$ (pairwise) scalings. Source: §10, §10c.

at 100% at all tested combinations of $(\sigma_s, \sigma_r, n_{\text{tracks}})$; false rate tracks the combinatorial expectation. The paper-style 2×2 figures for publication are in `outputs/segment_analysis/paper/fig13{a--e}.pdf`.

4 Solver and track-level results

4.1 Solver segment efficiency (§14)

Feeding the accepted triplets through the sparse Hopfield Hamiltonian and thresholding at $\tau = 0.35$ yields the four-panel Fig. 5: segment efficiency is flat at 100% across $n_{\text{tracks}} \in [2, 1000]$. The false-rate (fraction of accepted segments that are false) climbs from 0 at low n_{tracks} to $(19.8 \pm 0.4)\%$ at $n_{\text{tracks}} = 1000$. The raw pair counts in panel (iii) again recover the n (true) and n^3 (false) slopes; the post-threshold counts in panel (iv) collapse the false count by several orders of magnitude at low n_{tracks} — but at $n_{\text{tracks}} = 1000$ the false-active count approaches 10^3 per event.

4.2 Track-level performance and the knee (§15)

Figure 6 shows the six-panel track-level summary from `EventValidator`. Reconstruction efficiency plateaus at $\approx 99\%$ up to $n_{\text{tracks}} = 500$, drops to $\approx 85\%$ at $n_{\text{tracks}} = 750$, and collapses to $(44 \pm 3)\%$ at $n_{\text{tracks}} = 1000$. The ghost rate peaks at $\approx 3.3\%$ near $n_{\text{tracks}} = 500$–750 then *falls* at $n_{\text{tracks}} = 1000$ — not because ghosts vanish, but because the connected-component tracker absorbs false-segment bridges into mega-clusters which fail the purity cut and therefore neither count as matched nor as ghost-reco. Clone fraction is numerically zero. Mean purity of accepted tracks remains $= 1.00$; mean hit efficiency is 1.000 for clean events and (0.996 ± 0.002) under a 1% hit-drop model, consistent with the structural loss expected from dropping on average one hit per five.

4.3 Spectral diagnostics (§16)

To determine whether the efficiency collapse originates in the solver, the Hamiltonian matrix is analysed by sparse ARPACK eigensolve [9]. The results are summarised in Table 1 and Fig. 7. The minimum eigenvalue bottoms out at $\lambda_{\min} = 0.88$ for $n_{\text{tracks}} \geq 750$; the maximum eigenvalue saturates at $\lambda_{\max} \approx 7.1$; the condition number remains bounded by $\kappa \lesssim 9$. The Gershgorin lower bound $\min_i(d_i - r_i)$ saturates at zero for $n_{\text{tracks}} \geq 500$, indicating that the coupling mass per

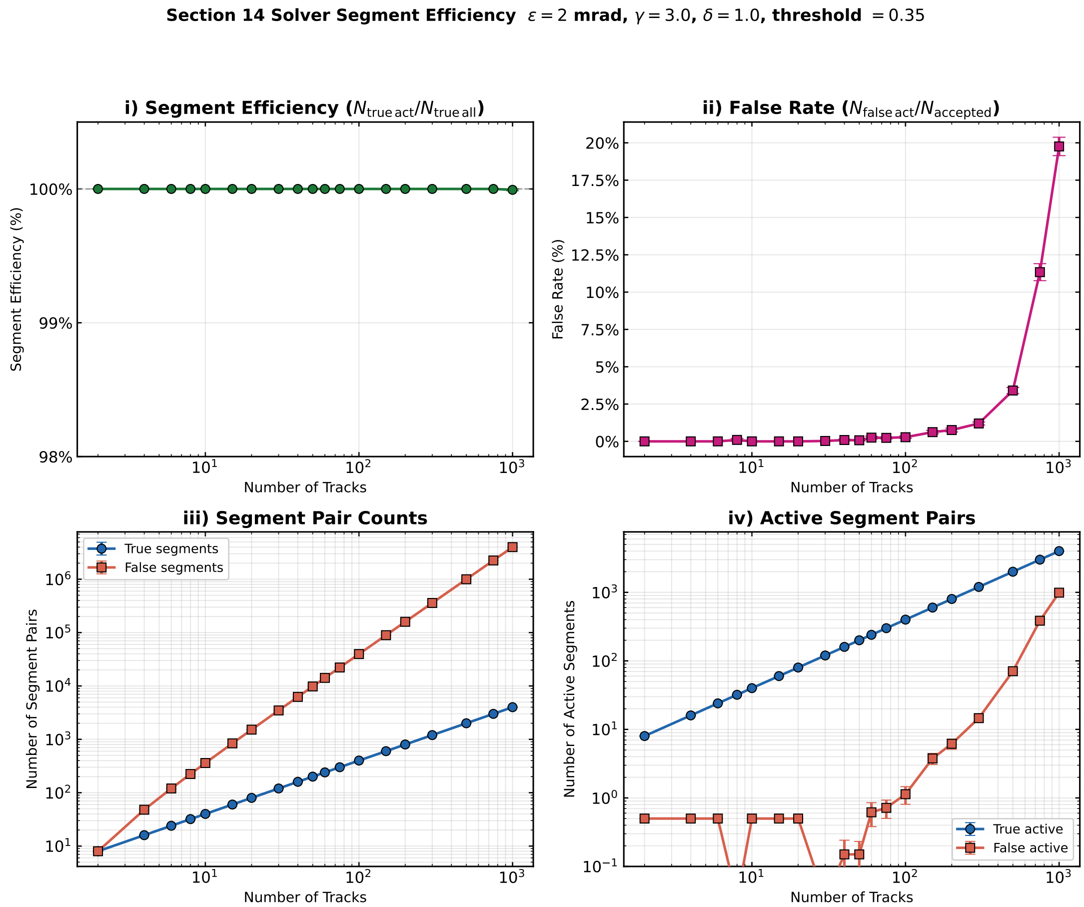

Figure 5: §14: Solver segment efficiency (i), false rate (ii), raw pair counts (iii), and post-threshold active counts (iv) versus multiplicity. $\gamma = 3$, $\delta = 1$, $\varepsilon = 2$ mrad, $\tau = 0.35$.

row matches the diagonal, but ARPACK confirms the actual spectrum stays strictly positive: the matrix is everywhere invertible and the SPD direct solve is numerically healthy.

4.4 Solution histograms (§16c)

The per-segment solution distribution (Fig. 8) shows a tight false-segment peak at the isolated-Hopfield fixed point $s^*_{\mathrm{fp}} = \delta/(\delta + \gamma) = 0.25$ and a broader true-segment population with outer-chain invariant $s^*_{\mathrm{outer}} = 0.3636$. At all tested multiplicities the threshold $\tau = 0.35$ separates the peaks cleanly. A small high-s false-tail emerges at large n_{tracks} (Table 2): 0, 9, 32, 306, 1119, 2788 false activations at $n_{\mathrm{tracks}} \in \{10, 100, 300, 500, 750, 1000\}$, translating to a false-rate of 0.00%–0.02% of the total false pool — which is nonetheless a large *absolute* number at high multiplicity because the false pool itself scales as n^3_{tracks}.

4.5 Tracker A/B (§17)

The same set of solver-active segments is fed into two trackers (Table 3, Fig. 9). The connected-component tracker (`get_tracks`) and the layered tracker (`get_tracks_layered`) share the same solver output but differ dramatically in their high-n_{tracks} behaviour: CC efficiency collapses from 98.8% at $n_{\mathrm{tracks}} = 100$ to 42.8% at $n_{\mathrm{tracks}} = 1000$, whereas the layered tracker remains at $\geq 99.96\%$ across the same range. The layered tracker pays with a ghost rate that grows from 0.1% to 23.6% over the same span, produced almost entirely by the small-but-rising high-s false tail of §4.

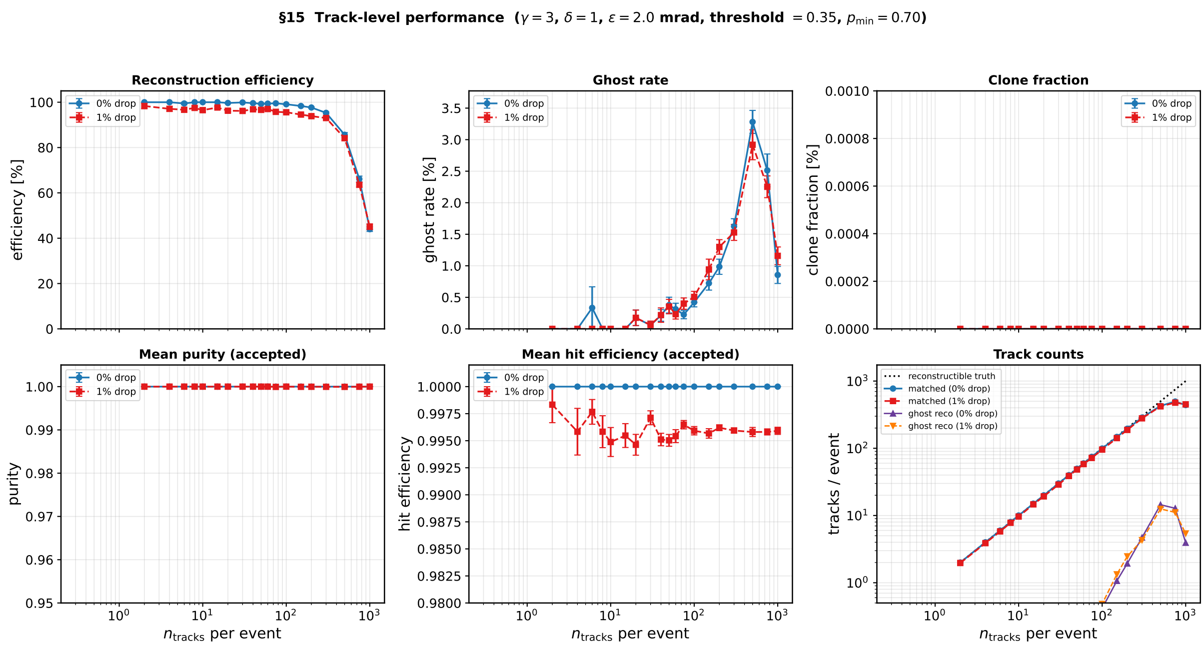

Figure 6: §15: track-level metrics under `EventValidator.match_tracks` with purity ≥ 0.70, `min_rec_hits`$= 3$, for clean and 1% hit-drop events. The efficiency knee appears between $n_{\text{tracks}} = 500$ and $n_{\text{tracks}} = 1000$.

Table 1: Spectral quantities of the Hopfield Hamiltonian at fixed $(\gamma, \delta, \varepsilon) = (3, 1, 2\,\text{mrad})$.

n_{tracks}	$\lambda_{\min}$	$\lambda_{\max}$	κ	$\max_i r_i$
10	2.38	5.62	2.36	2.0
100	2.22	5.80	2.61	2.7
300	1.98	6.00	3.03	3.0
500	1.02	6.94	7.5	4.0
750	0.88	7.12	8.6	4.0
1000	0.88	7.12	8.2	4.0

5 Discussion

The results in §3 demonstrate that the analytic threshold $\varepsilon(\sigma_s, \sigma_r)$ is *overtight* only by the safety factor $s = 3$ it was constructed with: true-pair acceptance stays at 100% across the full sweep grid and the observed false-rate climb is entirely due to the combinatorial growth of the false-pair pool, not to a mismatched ε. The triplet/pairwise comparison in §3.4 recovers exactly the expected T^3 and T^2 combinatorial scalings, and so the new package reproduces the older reference to within statistical error — the only genuine difference between them is the choice of segment-pair definition.

The high-multiplicity behaviour identified in §4 is the central finding of this study. The solver itself remains healthy: the Hamiltonian spectrum is strictly positive, the matrix is sparse and well-conditioned ($\kappa \leq 9$), and the per-segment solution populations show the expected Hopfield fixed points $s^*_{\text{fp}} = 0.25$ and $s^*_{\text{outer}} = 0.3636$ with threshold-preserving separation at all tested n_{tracks}. The false-rate growth at high n_{tracks} is therefore structural, not numerical: with $\mathcal{O}(n^3_{\text{tracks}})$ false-pair candidates, even an $\mathcal{O}(10^{-4})$ post-threshold misclassification rate produces $\mathcal{O}(10^3)$ ghost segments per event at $n_{\text{tracks}} = 1000$.

The resulting tracker failure (Fig. 9) is a post-processing artefact: the connected-component algorithm treats the false-segment "bridges" as legitimate edges and merges entire bundles of

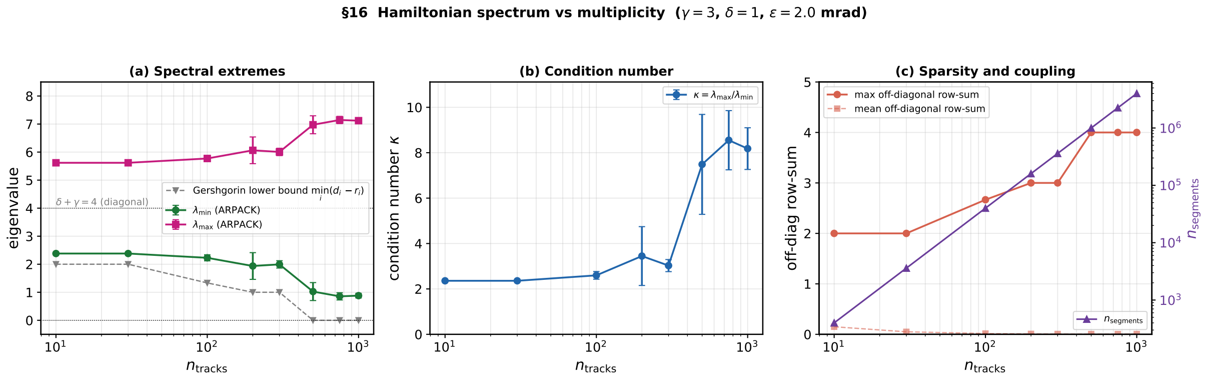

Figure 7: §16: extremal eigenvalues, condition number, and off-diagonal row-sum versus multiplicity.

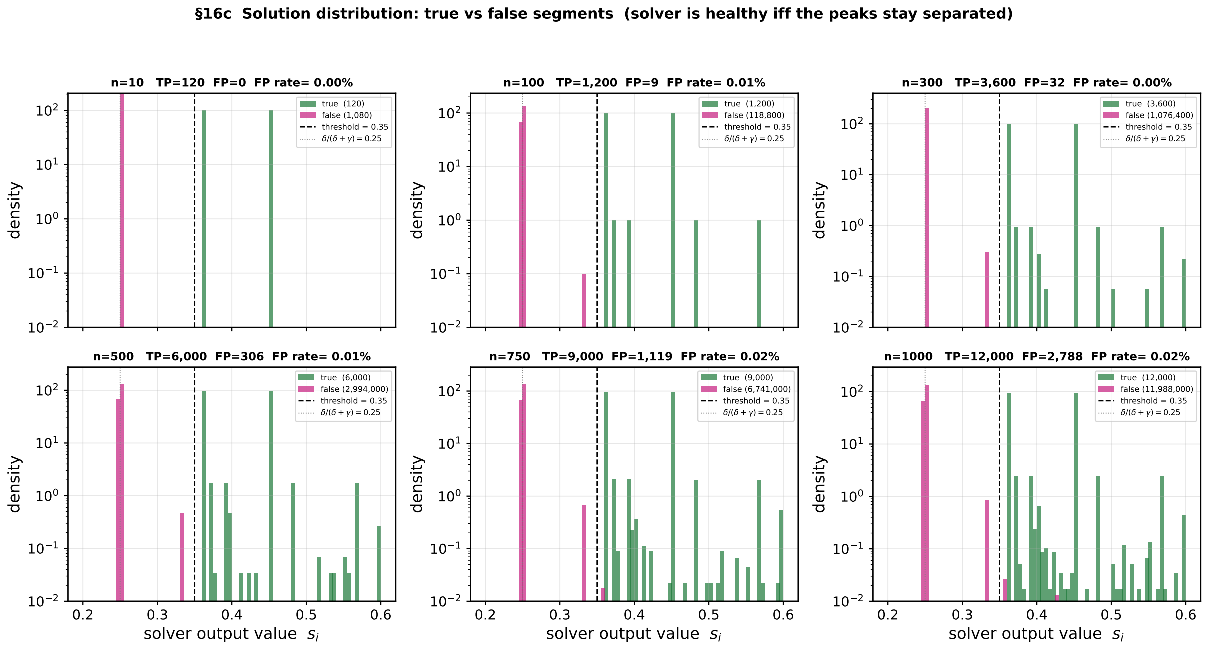

Figure 8: §16c: solver output distributions for true (green) and false (magenta) segments. Dashed: threshold $\tau = 0.35$; dotted: $s^*_{\mathrm{fp}} = 0.25$.

real tracks into unmatchable super-clusters; the layered tracker, by contrast, imposes module-exclusivity and angular consistency, rejects bridging segments, and recovers $\geq 99.9\%$ reconstruction efficiency — at the price of a 23.6% ghost rate at $n_{\mathrm{tracks}} = 1000$. The validated Hamiltonian operating point $(\gamma, \delta, \varepsilon, \tau) = (3,\ 1,\ 2\,\mathrm{mrad},\ 0.35)$ therefore supports up to $n_{\mathrm{tracks}} = 10^3$ tracks per event with either tracker, at opposite ends of the efficiency-vs-purity trade-off.

Limitations.

- The spectrum scan (§16) uses a single event per multiplicity point; the error bars on $\lambda_{\max}$ at high n_{tracks} are indicative only.
- Only a 1% flat hit-drop model is tested; no ghost-hit or module-inefficiency models are included.
- The $s \approx 0.45$–0.6 false-tail at $n_{\mathrm{tracks}} = 1000$ (Fig. 8) is not characterised in detail; tightening τ or raising γ would suppress it at some cost to the true-peak lower tail.

Table 2: False-positive counts from §16c.

n_{tracks}	TP	FP $> \tau$	FP fraction
10	120	0	0.00%
100	1 200	9	0.01%
300	3 600	32	0.00%
500	6 000	306	0.01%
750	9 000	1 119	0.02%
1000	12 000	2 788	0.02%

Table 3: §17: tracker A/B comparison (clean events).

n_{tracks}	eff (CC)	eff (layered)	ghost (CC)	ghost (layered)
100	98.8%	100.0%	0.5%	0.1%
300	94.7%	100.0%	0.9%	1.2%
500	85.4%	100.0%	1.6%	3.9%
750	67.1%	99.96%	3.3%	11.7%
1000	42.8%	99.98%	0.9%	23.6%

6 Conclusion

The `lhcb_velo_toy` stack has been verified against the earlier reference implementation and characterised up to $n_{\text{tracks}} = 10^3$ tracks per event. Within the validated envelope $(\gamma, \delta, \varepsilon, \tau) =$ (3, 1, 2 mrad, 0.35), the Hamiltonian segment solver is numerically healthy and delivers 100% segment efficiency. The efficiency collapse observed with the default connected-components tracker is a *tracker* artefact and is fully reversed by a module-exclusive layered tracker.

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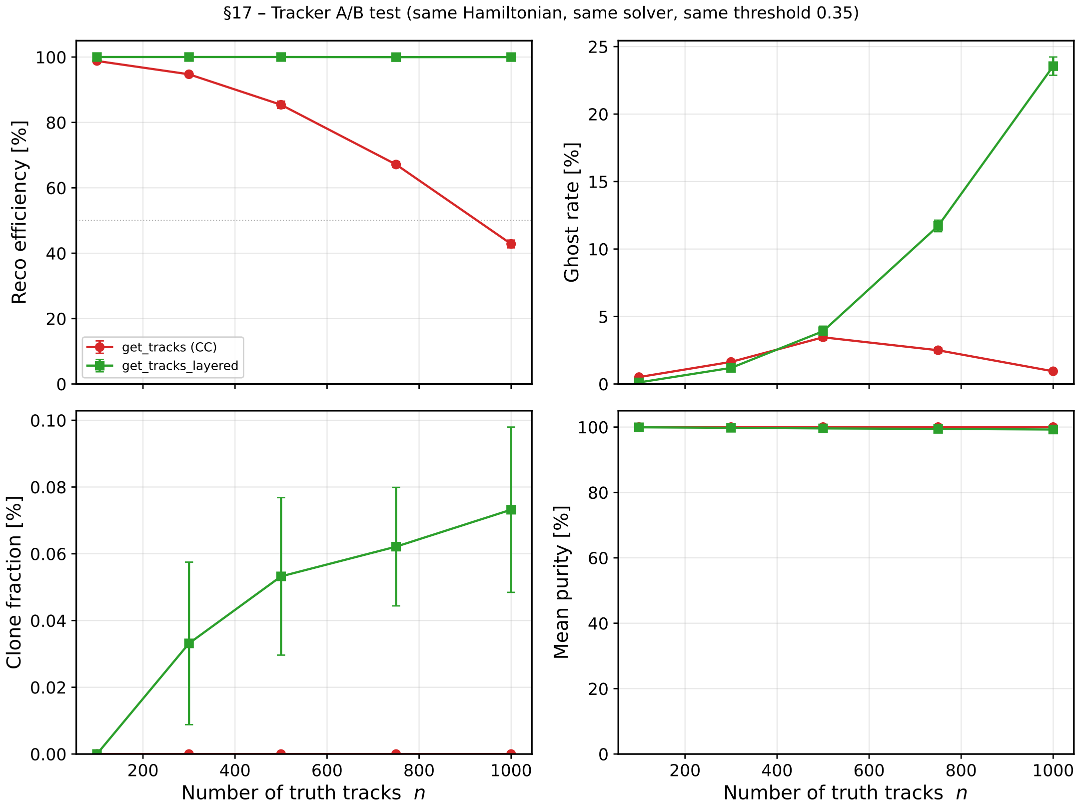

Figure 9: §17: reconstruction efficiency and ghost rate for the two trackers, same solver output.

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